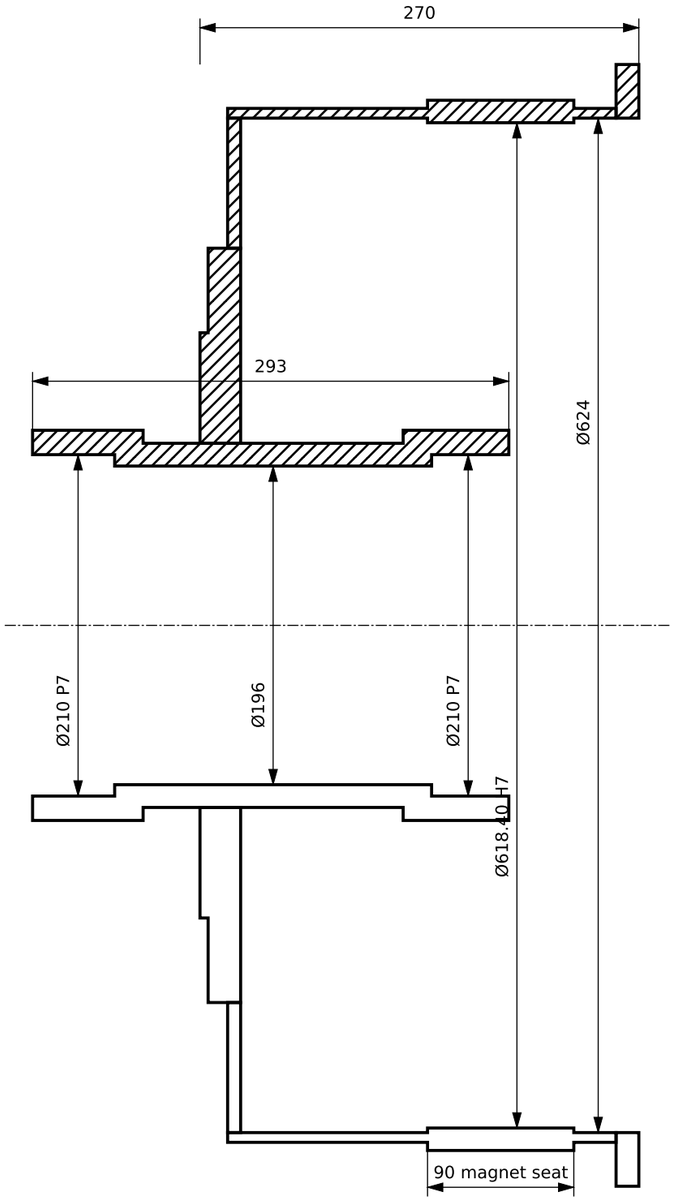


SECTION A-A (weld preps not shown, note 2)

Feature	Size	Tolerance / note
cup seats (2)	Ø210 × 45	P7, Ra 0.8, coax. 0.02
cup shoulders	Ø196	square 0.01 to axis
housing OD	Ø240 / Ø224	seal cover faces at x 92, 385
hub spigot	Ø360 × 5	f7; face flat 0.03
hub web	20 thick, Ø464	12× M16 thru PCD 400
outer web	8 thick	full pen. weld to shell
magnet seat	Ø618.40 × 90	H7, coax. 0.02 to cups
shell OD	Ø646 / Ø636	magnet zone 13.8 wall
shell ID (free)	Ø624	clearance to magnets
rear flange ring	Ø624/Ø690 × 14	M10 PCD 644, M12 PCD 662
lock holes	12× Ø26 H8 ↧20	PCD 662, interleaved

NOTES

1. Parts: housing tube (seamless S355J2H or ring forging), hub web 20 mm, outer web 8 mm, shell (rolled 16 mm plate, one longitudinal full-penetration weld, or ring rolled), rear flange ring 14 mm.
2. Welds: full penetration where the web meets the housing and the shell; ISO 5817 class B; 100 % MT, UT on the shell seam. Stress relieve 580 to 620 °C before machining.
3. Machine in one set-up after stress relief: cup seats Ø210 P7, magnet seat Ø618.40 H7, hub spigot Ø360 f7, rear flange face. Coaxiality of both cup seats and magnet seat ≤ 0.02; cup shoulder faces square to axis ≤ 0.01.
4. Hub web: 12× M16-6H through, PCD 400 (doc 13 joint, 10.9 preloaded). Rear flange ring: 24× M10-6H ↧20 PCD 644 (rear cover) and 24× M12-6H ↧20 PCD 662 + 12× Ø26 H8 ↧20 lock holes PCD 662 interleaved (36 positions at 10°).
5. Front and rear housing end faces: 6× M8 ↧16 on PCD 225 for the seal covers NP1-211; 2 grease nipples M10×1 into each bearing space with purge holes opposite.
6. Balance the finished rotor (drum, magnets, disc) to ISO 21940-11 G2.5 at 250 rpm, correction on the rear flange ring.
7. Magnet seat: degrease, grit blast Sa 2½ just before bonding; no paint inside. Outside: zinc spray + epoxy (C5) after machining.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

ROTOR DRUM WELDMENT (GENERATOR YOKE + BEARING HOUSING)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-210	A · 2026-11-02	1:3.8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2+N, welded, stress relieved	96.0 kg	Claude / not checked